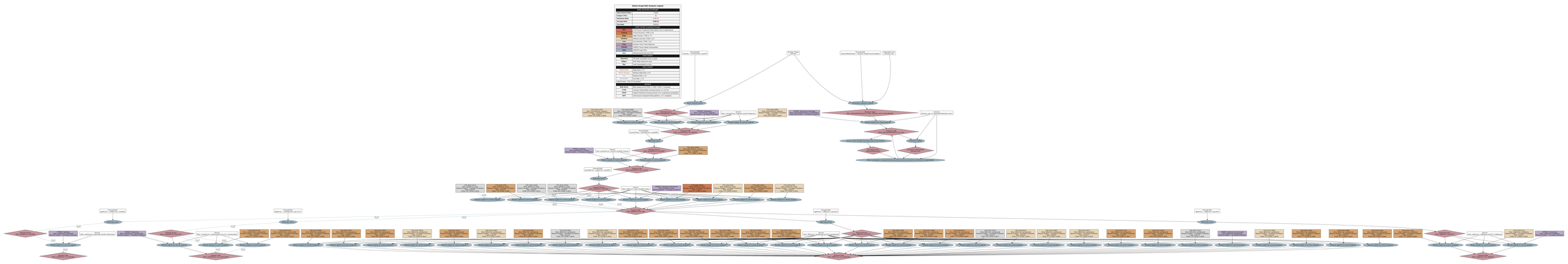

Attack Graph Risk Analysis Legend

RISK ANALYSIS SUMMARY	
Total Attack Paths:	14204
Unique CVEs:	45
Maximum Risk:	**0.00/10**
Average Risk:	**0.08/10**
Coverage:	100.0%
NODE COLORS (Colorblind-Friendly)	
KEV	CISA Known Exploited Vulnerability (active exploitation)
Critical	Critical Severity: CVSS ≥ 9.0
High	High Severity: CVSS ≥ 7.0
Medium	Medium Severity: CVSS ≥ 4.0
Low	Low Severity: CVSS < 4.0
Goal	Attacker Goal / Final Objective
STRIDE	STRIDE Threat Model Vulnerability
Rule	AND/OR Logic Rule
Fact	Infrastructure Fact (no CVE)
NODE SHAPES	
Diamond	OR Node (reachable state or goal)
Ellipse	AND Node (inference rule)
Box	Leaf (vulnerability or fact)
EDGE COLORS	
Dark Coral	High Risk ≥ 7.0
Burnt Orange	Medium-High Risk ≥ 4.0
Tan	Medium Risk ≥ 2.0
Steel Blue	Low Risk < 2.0
Label format: "risk: X.X (N paths)"	
METRICS	
Risk Score	Risk rating (0-10): EPSS × CVSS × KEV × Criticality
CVSS	Common Vulnerability Scoring System v3.1 (0-10)
EPSS	Exploit Prediction Scoring System (0-1): exploitation probability
KEV	CISA Known Exploited Vulnerabilities: 1.5× multiplier

Firewall Rule
internet → reverseProxy | tcp:443
Attacker Origin
internet
Firewall Rule
clinicianWorkstation → internet | httpProtocol:httpPort
Vulnerable User
clinician_user
Direct network access
Browsing a malicious website
CVE-2026-33555
Host: reverseProxy | haproxy
Remote Exploit → Privilege Escalation
Risk: 0.03/10
CVSS: 5.8 | EPSS: 0.0058
CVE-2026-26080
Host: reverseProxy | haproxy
Remote Exploit → Privilege Escalation
Risk: 0.02/10
CVSS: 3.7 | EPSS: 0.0055
Network Access
Host: reverseProxy | tcp:443
STRIDE: Tampering
Host: reverseProxy | haproxy
Remote Exploit → Privilege Escalation
Service
Host: reverseProxy | haproxy tcp:443 (haproxy)
CVE-2026-26081
Host: reverseProxy | haproxy
Remote Exploit → Privilege Escalation
Risk: 0.03/10
CVSS: 4.8 | EPSS: 0.0048
STRIDE: Elevation of privilege
Host: clinicianWorkstation | thunderbird
Client-Side Exploit → Privilege Escalation
Malicious Input
Host: clinicianWorkstation | victim: clinician_user via thunderbird
Account
clinician_user on clinicianWorkstation (user)
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit for a client program
Firewall Rule
reverseProxy → portalServer | tcp:8080
Execute Code
Host: reverseProxy | as: haproxy
Execute Code
Host: clinicianWorkstation | as: user
Multi-hop access
Access a host through executing code on the machine
Password sniffing
STRIDE: Spoofing
Host: portalServer | tomcat
Remote Exploit → Privilege Escalation
Service
Host: portalServer | tomcat tcp:8080 (tomcat)
Network Access
Host: portalServer | tcp:8080
CVE-2023-24998
Host: portalServer | tomcat
Remote Exploit → Privilege Escalation
Risk: 3.66/10
CVSS: 7.5 | EPSS: 0.4879
Can Access Host
clinicianWorkstation
Principal Compromised
clinician_user
Remote exploit of a server program
Remote exploit of a server program
When a principal is compromised any machine he has an account on will also be compromised
Firewall Rule
portalServer → apiServer | tcp:8443
Execute Code
Host: portalServer | as: tomcat
Multi-hop access
CVE-2026-12611
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.00/10
CVSS: 0.0 | EPSS: 0.0025
CVE-2026-5795
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.07/10
CVSS: 7.4 | EPSS: 0.0061
CVE-2026-19204
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.00/10
CVSS: 0.0 | EPSS: 0.0031
CVE-2026-19203
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.00/10
CVSS: 0.0 | EPSS: 0.0031
Network Access
Host: apiServer | tcp:8443
Service
Host: apiServer | jetty tcp:8443 (jetty)
STRIDE: Information disclosure
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
CVE-2026-10050
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 9.1 | EPSS: 0.0063
CVE-2026-8384
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.03/10
CVSS: 5.3 | EPSS: 0.0033
CVE-2026-10051
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.03/10
CVSS: 7.5 | EPSS: 0.0030
CVE-2026-6790
Host: apiServer | jetty
Remote Exploit → Privilege Escalation
Risk: 0.02/10
CVSS: 5.3 | EPSS: 0.0031
risk: 0.00
(4 paths)
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
risk: 0.00
(3 paths)
risk: 0.00
(3 paths)
risk: 0.00
(4 paths)
Execute Code
Host: apiServer | as: jetty
Firewall Rule
apiServer → authServer | tcp:9443
Firewall Rule
apiServer → cacheServer | tcp:11211
Firewall Rule
apiServer → dbServer | tcp:5432
Firewall Rule
apiServer → mqServer | tcp:5672
risk: 0.00
(3 paths)
risk: 0.00
(4 paths)
Multi-hop access
Multi-hop access
Multi-hop access
Multi-hop access
risk: 0.00
(3 paths)
risk: 0.00
(4 paths)
Network Access
Host: authServer | tcp:9443
STRIDE: Spoofing
Host: authServer | keycloak
Remote Exploit → Privilege Escalation
Service
Host: authServer | keycloak tcp:9443 (keycloak)
STRIDE: Tampering
Host: cacheServer | memcached
Remote Exploit → Privilege Escalation
Network Access
Host: cacheServer | tcp:11211
Service
Host: cacheServer | memcached tcp:11211 (memcached)
CVE-2026-47784
Host: cacheServer | memcached
Remote Exploit → Privilege Escalation
Risk: 0.04/10
CVSS: 8.1 | EPSS: 0.0055
CVE-2026-47783
Host: cacheServer | memcached
Remote Exploit → Privilege Escalation
Risk: 0.10/10
CVSS: 8.1 | EPSS: 0.0126
CVE-2026-14669
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.12/10
CVSS: 8.8 | EPSS: 0.0066
CVE-2026-14677
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0042
CVE-2026-14670
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0044
CVE-2026-18024
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 4.3 | EPSS: 0.0019
CVE-2026-14671
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0042
CVE-2026-14663
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 6.5 | EPSS: 0.0010
CVE-2026-6464
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.06/10
CVSS: 8.1 | EPSS: 0.0036
CVE-2026-6469
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 3.8 | EPSS: 0.0032
CVE-2026-14678
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 4.3 | EPSS: 0.0019
CVE-2026-15742
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.10/10
CVSS: 8.8 | EPSS: 0.0056
CVE-2026-14676
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0044
CVE-2026-16238
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0041
CVE-2026-14664
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0044
CVE-2026-18408
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.07/10
CVSS: 8.8 | EPSS: 0.0036
CVE-2026-14679
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.03/10
CVSS: 8.2 | EPSS: 0.0029
Service
Host: dbServer | postgresql tcp:5432 (postgresql)
Network Access
Host: dbServer | tcp:5432
CVE-2026-19385
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0043
CVE-2026-14680
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 8.8 | EPSS: 0.0042
CVE-2026-6471
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.08/10
CVSS: 7.2 | EPSS: 0.0053
CVE-2026-14673
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 3.8 | EPSS: 0.0017
CVE-2026-6470
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 4.3 | EPSS: 0.0033
CVE-2026-14681
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.00/10
CVSS: 4.2 | EPSS: 0.0007
CVE-2026-14672
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 5.3 | EPSS: 0.0025
CVE-2026-15741
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.09/10
CVSS: 8.8 | EPSS: 0.0047
CVE-2026-16239
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.13/10
CVSS: 8.8 | EPSS: 0.0068
CVE-2026-16241
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 3.8 | EPSS: 0.0020
STRIDE: Information disclosure
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
CVE-2026-14666
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.01/10
CVSS: 4.2 | EPSS: 0.0018
CVE-2020-13692
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.59/10
CVSS: 7.7 | EPSS: 0.0409
CVE-2026-14668
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.04/10
CVSS: 8.1 | EPSS: 0.0033
CVE-2026-14662
Host: dbServer | postgresql
Remote Exploit → Privilege Escalation
Risk: 0.09/10
CVSS: 8.8 | EPSS: 0.0046
CVE-2019-11287
Host: mqServer | rabbitmq
Remote Exploit → Privilege Escalation
Risk: 0.25/10
CVSS: 7.5 | EPSS: 0.0440
Network Access
Host: mqServer | tcp:5672
Service
Host: mqServer | rabbitmq tcp:5672 (rabbitmq)
CVE-2020-5419
Host: mqServer | rabbitmq
Remote Exploit → Privilege Escalation
Risk: 0.05/10
CVSS: 6.7 | EPSS: 0.0045
STRIDE: Tampering
Host: mqServer | rabbitmq
Remote Exploit → Privilege Escalation
Remote exploit of a server program
risk: 0.00
(3 paths)
Remote exploit of a server program
Remote exploit of a server program
Remote exploit of a server program
risk: 0.00
(3 paths)
Execute Code
Host: authServer | as: keycloak
Execute Code
Host: cacheServer | as: memcached
Execute Code
Host: dbServer | as: postgresql
Execute Code
Host: mqServer | as: rabbitmq